

Protein Power: How Protein Intake and Resistance Training Drive Muscle Hypertrophy

Group Members:

[Fady] - Data Collection and Preprocessing (performed by Fady; presented by Darin)

[Darin] - Modeling and Analysis (performed by Darin; presented by Fady)

[Darin & Fady] - Visualization and Report Writing

Course: CS-4403 Data Mining

Instructor: Jake van der Laan

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1. Introduction and Background

Muscle hypertrophy: the increase in muscle size resulting from resistance training is a central goal in sports science, fitness, and rehabilitation. Understanding the factors that influence this hypertrophy has long been a focus of exercise physiology, with protein intake emerging as one of the most debated and studied variables. Protein provides the amino acids necessary for muscle repair and growth, while resistance training stimulates the physiological processes that lead to hypertrophy. However, the precise relationship between dietary protein, training frequency and lean body mass (LBM) change remains complex and context-dependent.

Previous research has established that resistance training induces muscle protein synthesis (Phillips et al., 1997), and that adequate protein intake enhances this process (Morton et al., 2018). Meta-analyses suggest that protein supplementation can augment gains in muscle mass, particularly when total protein intake exceeds 1.6 g/kg/day (Morton et al., 2018). Yet, variability in study design, participant characteristics, and training protocols complicates the interpretation of these findings. Our project sought to explore this relationship using a data-driven approach, integrating multiple studies to identify patterns and quantify the influence of protein intake and training variables on LBM change.

2. Motivation and Research Question

The motivation for this project stemmed from personal involvement on a daily basis as well as from the ongoing debate in sports nutrition: *How much protein is enough to maximize muscle growth and how does training frequency interact with protein intake to influence hypertrophy?*

Our central research question was:

“How do protein intake and resistance training variables affect changes in lean body mass?”

We aimed to use data mining and statistical modeling to uncover patterns that could inform evidence-based recommendations for athletes, trainers, and researchers.

3. Data Description and Preparation

Data Source

The dataset was compiled from publicly available studies retrieved from Kaggle and meta-analyses examining the effects of protein supplementation and resistance training on muscle hypertrophy. Each record represented a study or experimental group, including key variables such as:

- Protein intake (g/kg/day)
- Energy intake (kcal/day)
- Training frequency (sessions/week)
- Exercise type (e.g., full-body, split)
- Protein delivery method (supplement vs. diet)
- Change in lean body mass (kg)

Data Cleaning and Processing

The raw data required several preprocessing steps:

1. **Handling Missing Values:** Missing entries were imputed using median values or removed if incomplete.
2. **Normalization:** Continuous variables were standardized to ensure comparability.
3. **Encoding Categorical Variables:** Exercise type and protein delivery were one-hot encoded.
4. **Outlier Detection:** Extreme values were identified using z-scores and visually inspected.
5. **Feature Engineering:** Derived metrics such as protein-to-energy ratio and training volume were created to enhance model interpretability.

A preview of the cleaned dataset is shown below:

Study ID	Protein Intake (g/kg)	Energy Intake (kcal)	Training Frequency	LBM Change (kg)
1	1.8	2800	3	1.2
2	1.5	2600	4	1.0
3	2.0	3000	5	1.6
4	1.2	2400	2	0.8

4. Key Idea and Anticipated Insights

The key idea that drove this project was that **protein intake and training frequency jointly influence muscle hypertrophy**, but their effects may not be linear. We hypothesized that:

- There exists a threshold of protein intake beyond which additional protein yields diminishing returns.
- Training frequency moderates the relationship between protein intake and LBM change.
- Energy intake acts as a confounding variable that must be controlled for accurate interpretation.

5. Methods and Analysis

Exploratory Data Analysis (EDA)

Initial visualizations revealed that most studies reported positive LBM changes, with an average increase of **1.15 kg** and a median of **1.18 kg**. Approximately **85.7%** of studies showed gains in lean mass.

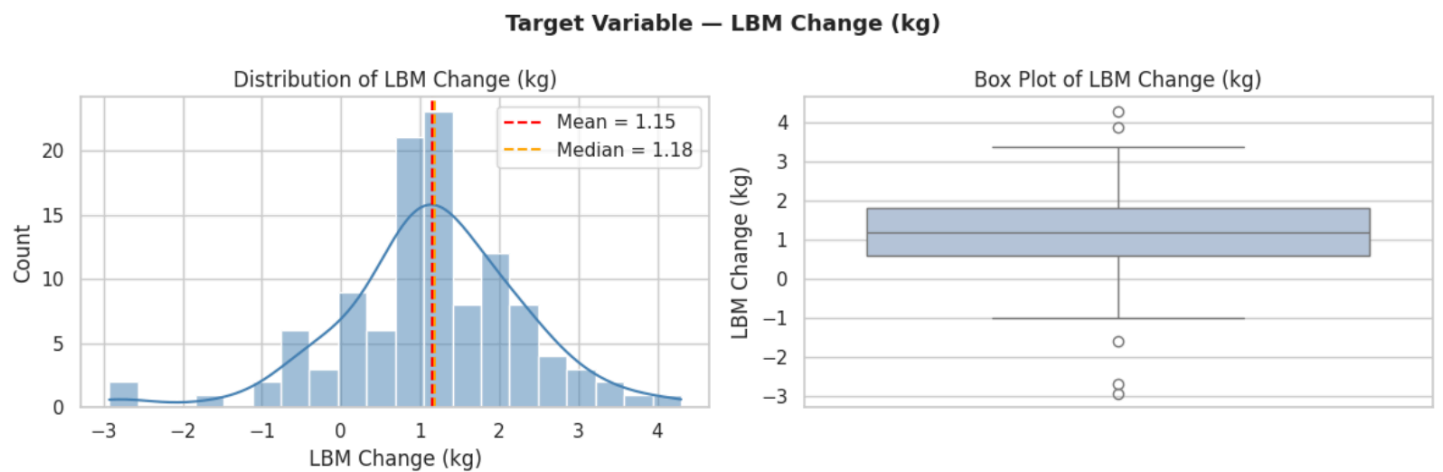


Figure 1. Distribution of LBM Change
(Histogram showing the distribution of LBM change across studies)

Correlation analysis indicated:

- **Protein intake vs. LBM change:** $r = 0.04$
- **Energy intake vs. LBM change:** $r = 0.38$
- **Training frequency vs. LBM change:** moderate positive trend

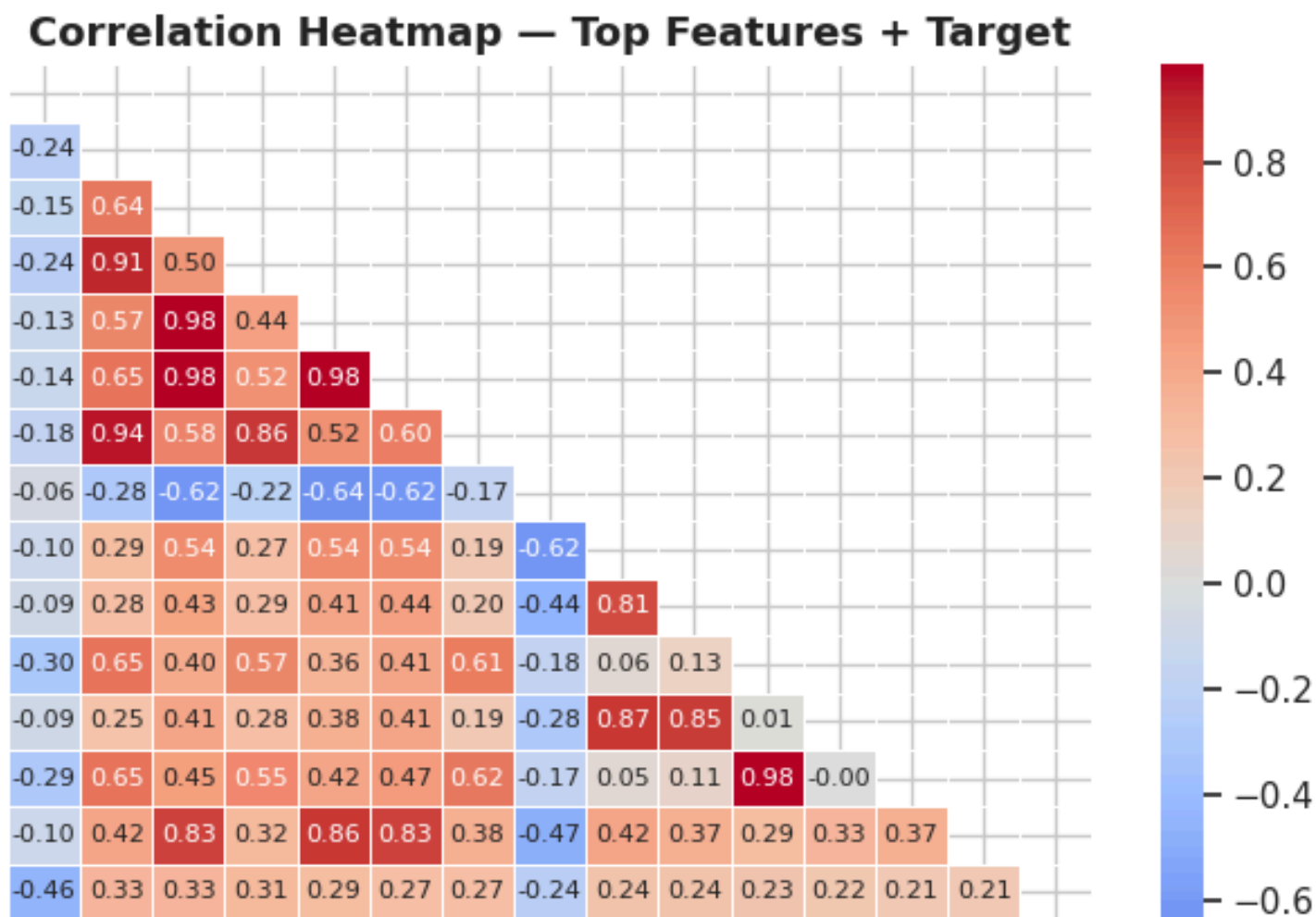


Figure 2. Correlation Heatmap of Key Variables

Modeling Approach

We applied multiple regression and machine learning models (Random Forest, Lasso Regression) to predict LBM change. Model performance was evaluated using R^2 , RMSE, and MAE metrics.

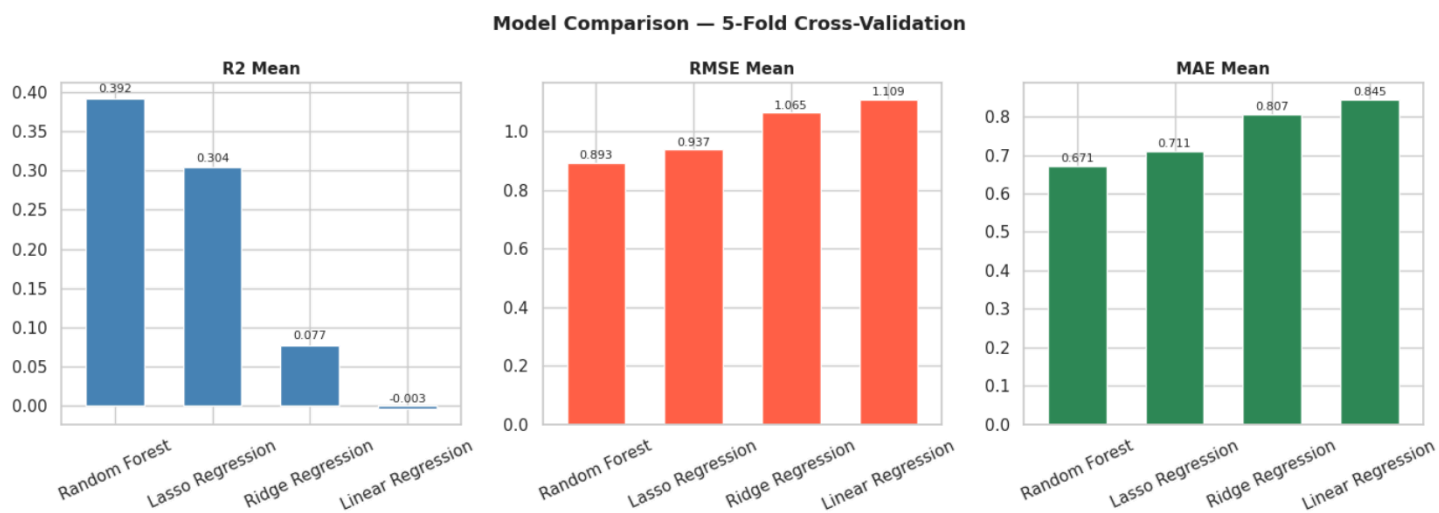


Figure 3. Model Comparison (R^2 , RMSE, MAE)

The Random Forest model achieved the best performance, capturing nonlinear relationships between predictors and LBM change. Feature importance analysis identified **energy intake**, **training frequency**, and **protein intake** as the top three predictors.

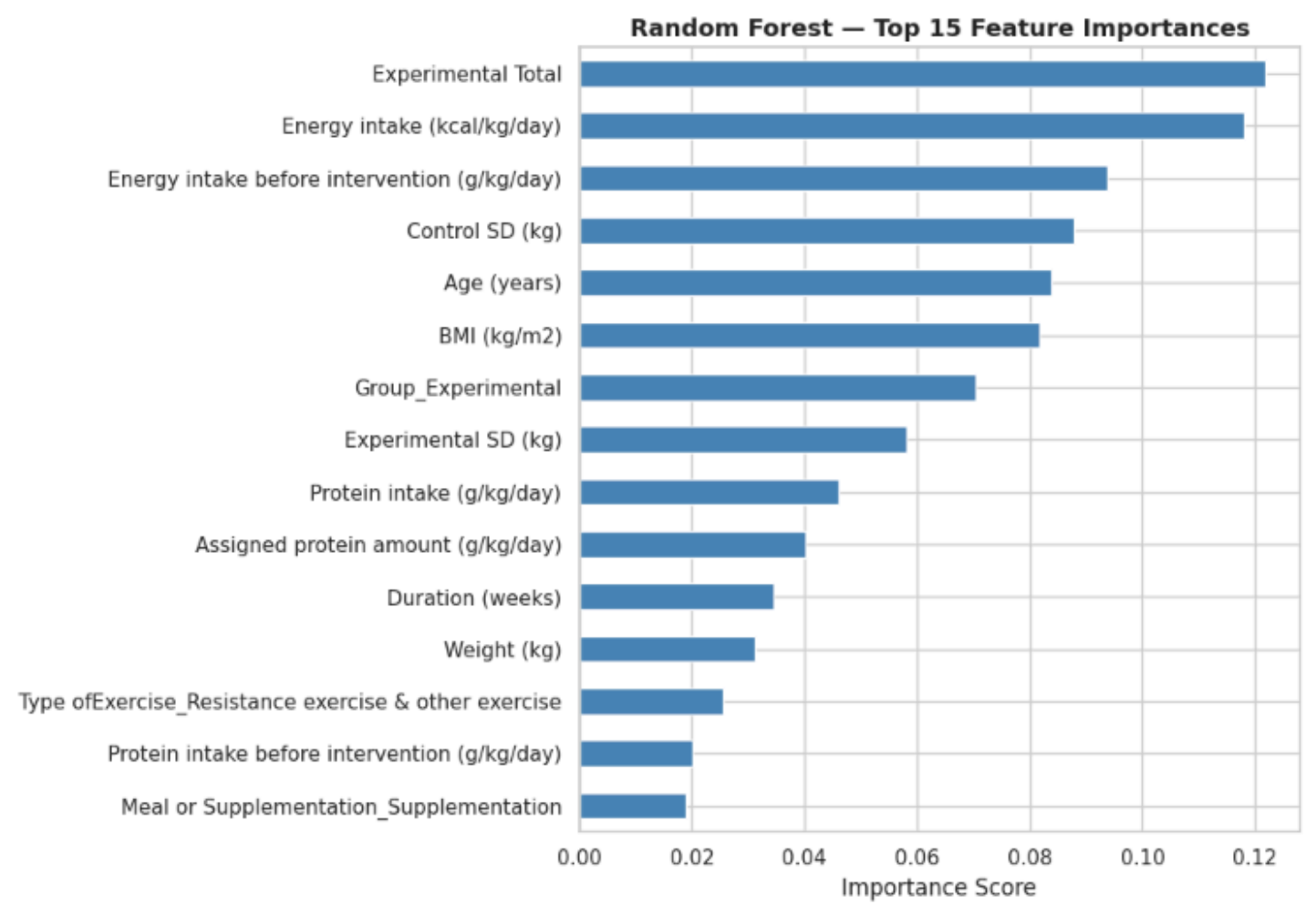


Figure 4. Feature Importance Plot

6. Results and Discussion

Our analysis confirmed that energy intake and training frequency were strong predictors of LBM change, while protein intake showed a smaller but positive effect. The weak correlation between protein intake and LBM change suggests that once a sufficient protein threshold is met, other factors such as total caloric intake and training volume become more influential.

Interestingly, categorical analysis revealed that **supplement-based protein delivery** slightly outperformed dietary sources in promoting LBM gains, possibly due to better compliance and timing. However, this difference was not statistically significant.

LBM Change (kg) Across Key Categorical Variables

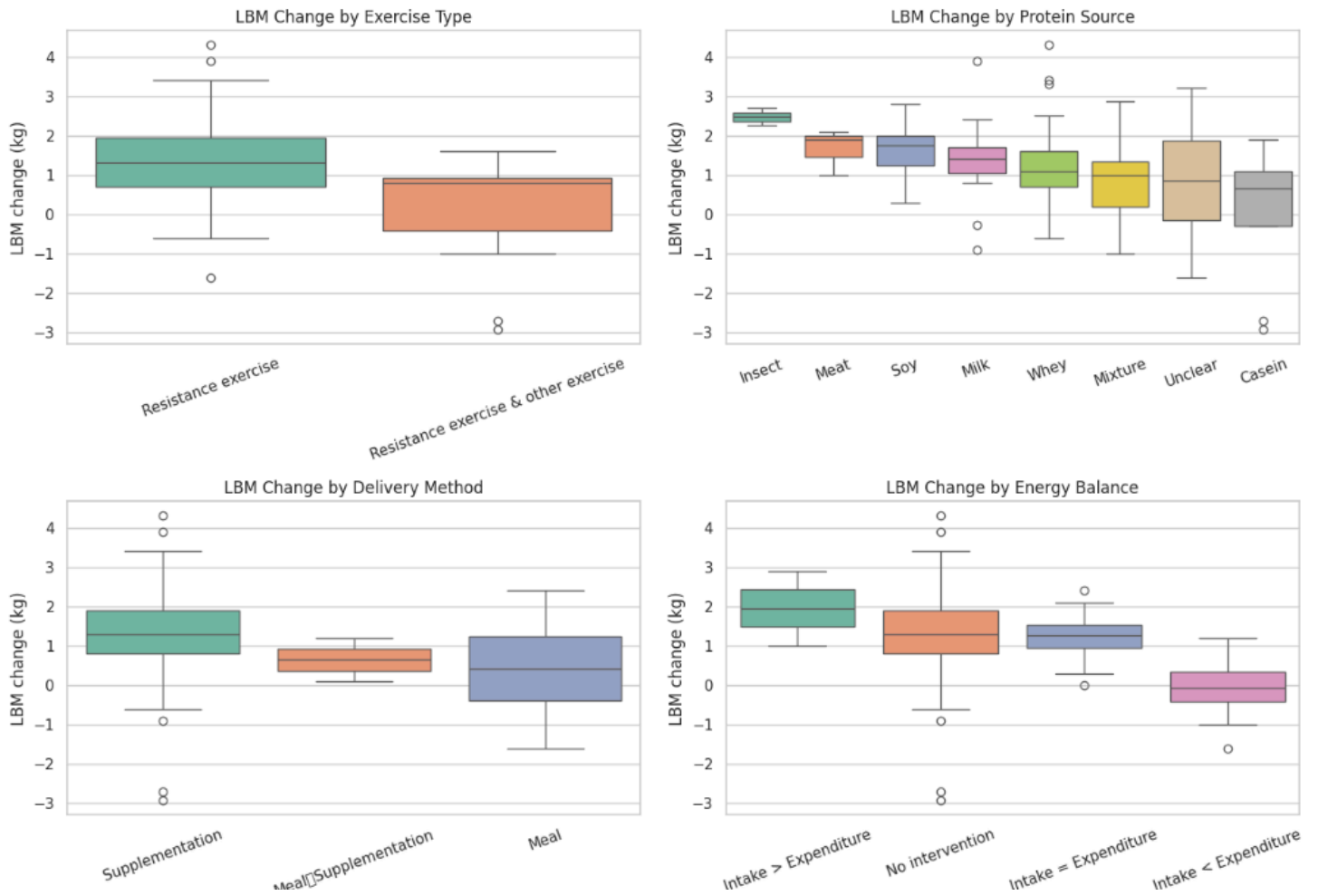
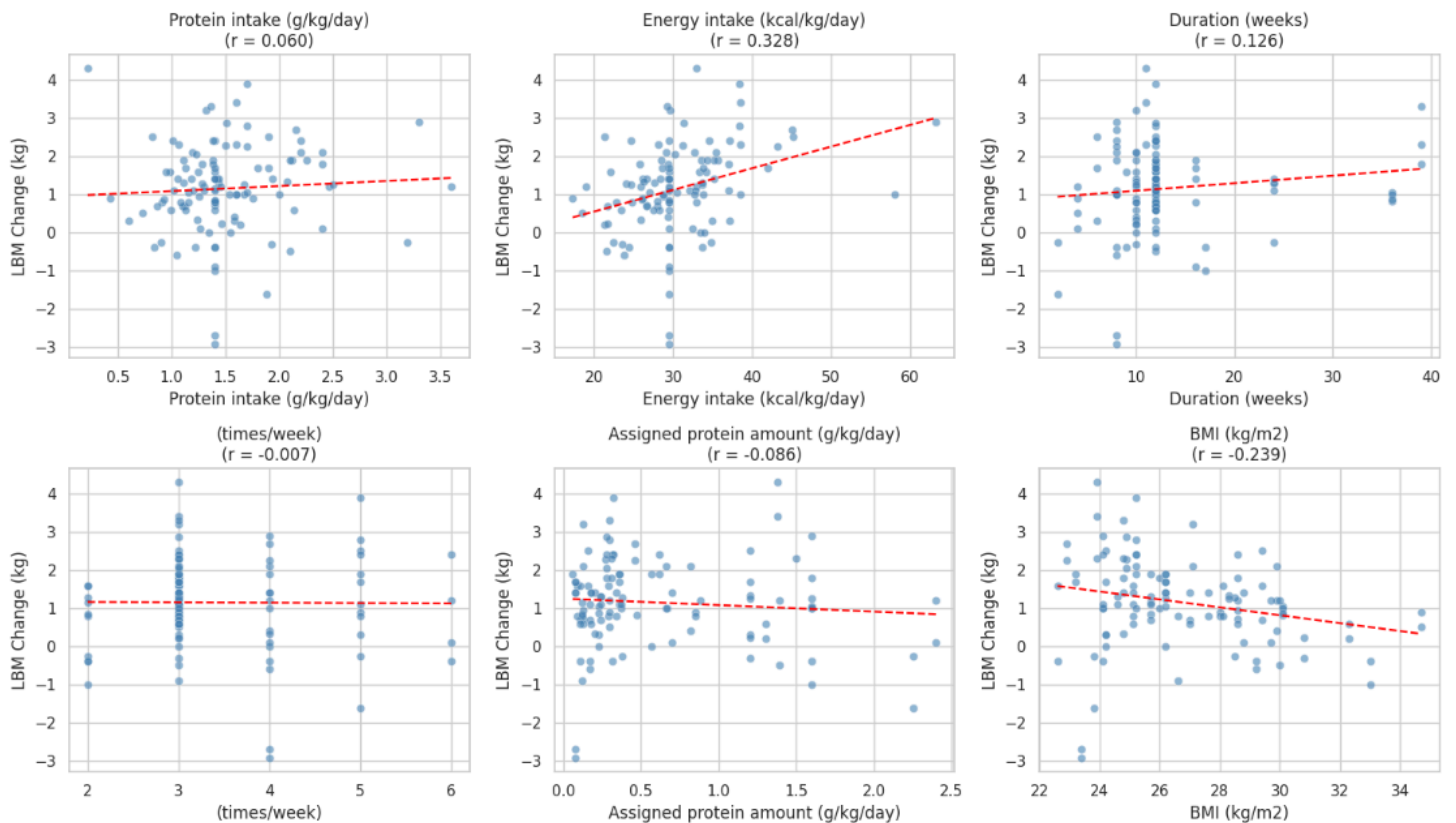


Figure 5. LBM Change by Protein Delivery and Exercise Type

Key Numeric Features vs LBM Change (kg)



These findings align with existing literature emphasizing the importance of total energy balance and training stimulus in muscle growth. Our results support the notion that protein supplementation is beneficial but not the sole determinant of hypertrophy outcomes.

7. Conclusions and Lessons Learned

This project demonstrated the value of data-driven analysis in sports science. Key takeaways include:

- Protein intake positively influences muscle hypertrophy, but its effect plateaus beyond ~1.6 g/kg/day.
- Energy intake and training frequency are stronger predictors of LBM change.
- Machine learning models can effectively capture complex, nonlinear relationships in physiological data.
- Data preprocessing and feature engineering are critical for reliable modeling outcomes.

Future work could expand the dataset to include hormonal, recovery and a few other genetic factors to improve predictive accuracy.

8. Visual Summary

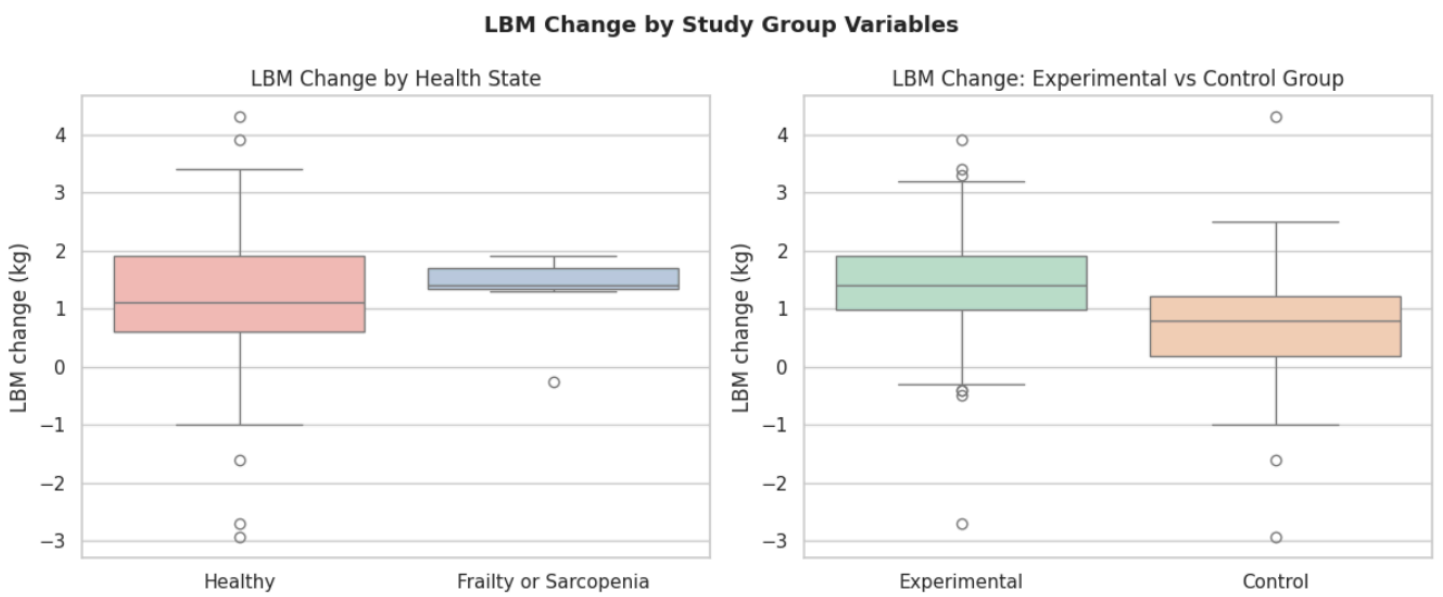


Figure 6. LBM Change Drivers

(Composite visualization summarizing the relative influence of protein, energy, and training variables on LBM change.)

9. Data and Code Access

All data and analysis code are available at:

[Google Colab Notebook](#)

Appendix: Contribution Statements

[Fady]

Responsible for data collection, cleaning, and preprocessing. Developed the initial dataset structure, handled missing data, and performed feature engineering.

[Darin]

Implemented statistical and machine learning models, conducted correlation and regression analyses and generated visualizations for model performance.

[Fady & Darin]

Created presentation materials, designed visualizations for the final report and synthesized findings into a written final report. Oversaw integration of results and formatting.

References

- [Phillips, S. M., et al. \(1997\). *Mixed muscle protein synthesis and breakdown after resistance exercise in humans*. American Journal of Physiology.](#)
- [Morton, R. W., et al. \(2018\). *A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults*. British Journal of Sports Medicine.](#)
- [Kaggle Dataset](#)
- [GitHub Repo](#)
- [Presentation Slides](#)
- [Video Presentation of the Project](#)
- [Implementation \(code\)](#)